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SUMMARY

I am a Software Engineer, with a strong passion for GNU/Linux, Unix systems, and Systems Programming, and a deep interest in their evolution from inception to the present. I am seeking a role in **Linux/Unix Systems Engineering or Programming** to contribute my skills while gaining hands-on experience in real-world projects.

EXPERIENCE

Platform Software Engineer **December 2025 - Now**
Sonnen GmbH

- Migrated multiple legacy cron jobs to **systemd** services and timers across sonnen's embedded Linux system, that integrate properly into **dpkg** lifecycle hooks, which improves logging and troubleshooting and centralizes the system management from systemd.
- Implemented **SoftAP** feature on sonnen's embedded Linux, bringing up a virtual access point over WiFi by orchestrating interface setup, dynamic config rendering for **hostapd** and **dnsmasq**, and iptables firewall rules, all managed through systemd units and a CLI tool.
- Optimized device configuration synchronization by transitioning from full payload uploads to incremental JSON **PATCH** updates, utilizing local caching and diffing to significantly reduce network and database overhead by **90%**.
- Redesigned Wi-Fi monitoring into an event-driven architecture by utilizing **NetworkManager** hooks and **hostapd_cli** action scripts to publish real-time interface state changes to an **MQTT** broker.
- Resolved a critical **single-radio hardware** limitation by automating **SoftAP** and station (STA) channel synchronization, utilizing a custom systemd timer-based monitor and NetworkManager dispatcher scripts with **exponential backoff**. Implementing exponential backoff in **NetworkManager** dispatcher scripts, that helped on resolving an hardware limitation.
- Resolved a critical fleet-wide false-offline issue by profiling the systemd boot sequence with **systemd-analyze** and modifying services, timers, and targets' definition to prevent blocking dependencies.
- Developed a **Wi-Fi watchdog** mechanism for the local system monitor to automatically detect and recover from USB enumeration errors, missing interfaces, Invalid configs, and driver-level crashes.
- Adapted a **3,000-line** legacy codebase to run simultaneously on Python **2.7** and Python **3.13** for a new platform, maintaining backward compatibility while resolving complex string **encoding/decoding** issues and replacing deprecated libraries

SKILLS & LANGUAGES

Technical skills & Tools:

- C, C++, Python, Java, OOP, and Data structures
- GNU/Linux, Bash, and docker
- Linux Network
- Git, GitHub, vim, and tmux

Languages:

- Arabic: Native proficiency
- English: Intermediate proficiency (B1)

PROJECTS

c-toolkit **July 2025 - September 2025**

- Developed reusable C99-based utilities, including:
  - **Arena allocator**: Developed thread-safe, region-based, and page-aligned memory allocator using **mmap**
  - **String**: Developed String utilities inspired by **Glib** from **Gnome Project**

BigNum **April 2025 - June 2025**

- Developed a C99 arbitrary-precision arithmetic library for cryptographic applications with architecture-dependent 32/64-bit word representation using arena allocation for memory management.
- Designed a robust API by exploring **OpenSSL** and **libtommath** libraries, and implemented core big number algorithms such as Euclidean division and modular inverse.

imglib **November 2024 - January 2025**

- Developed a lightweight, portable C library for image manipulation on Unix-like systems.
- Followed the **Suckless** C coding style and wrote a portable Makefile, avoiding GNU-specific extensions.
- Implemented robust error handling system inspired by **ffmpeg** and stride-based indexing for efficient pixel access.

Linux Kernel With lockdep **April 2024**

- Configured and built the Linux kernel to enable lockdep module which helps in debugging deadlocks in kernel space, then booted into the custom kernel to verify lockdep functionality.

CONTRIBUTIONS

- **Unikraft**:
  - Fixed broken documentation links in the main **README** and **sub-READMEs** in the examples directory for **unikraft/catalog** repo [#231] [#232]
  - Fixed a typo in the **ukallocregion** library's **allocregion_posix_memalign** function [#1708]
  - Working on updating **LLVM libraries** ports (**libcxx**, **libcxxabi**, **libunwind**, **libcompiler-rt**) from version **14.0.2** to **21.1.1** in the unikraft ecosystem, involving low-level build system work with **Makefiles**, **C++**, and **C**

EDUCATION

Faculty of Computers and Artificial Intelligence at Cairo University. **October 2021 - August 2025**

- Bachelor's Degree in Information Technology

CERTIFICATES

- CCNA: Introduction to Networks **July 2024 - August 2024**
- CCNA: Switching, Routing, and Wireless Essentials **July 2024 - August 2024**

EXTRACURRICULARS

IEEE Cairo University Student Branch **March 2024 - June 2024**

Networks Instructor

- Contributed to the development of technical content for the committee sessions and instructed real sessions for network committee members